

transition from concept to development. The founders focused on addressing a problem they viewed as largely overlooked, even as sectors such as fintech and e-commerce expanded rapidly.

The challenge was addressed over three years of in-house research and development, involving iterative prototyping and the development of adaptive power regulation algorithms. The system was built by an internal multidisciplinary team of around 19 members, including electronics engineers, power electronics specialists, software engineers working on energy and carbon-tracking systems, energy auditors, and deployment engineers. The company has secured patents in India and Germany and states that its systems comply with ISO standards for energy efficiency and emissions, as well as CE certification requirements.

The startup operates across three deployment models. OEM-embedded modules are integrated directly into devices at the design stage to eliminate standby losses at the source. Retrofit systems are deployed within existing infrastructure, such as offices, government buildings, and industrial facilities, via external switching and control units. Plug-and-play consumer products enable households to reduce energy waste through simple external adaptors for appliances such as chargers, laptops, and other electrical gadgets.

The company reports deploying



Saurabh Tiwari, CEO, Inno 91 Green Technologies

over 200 units across industrial, enterprise, and consumer environments, including installations in heavy industrial setups, such as railway workshop systems associated with IRCTC. These deployments have demonstrated measurable reductions in energy consumption, along with improvements in electrical efficiency and equipment lifespan. However, detailed financial performance figures have not been disclosed.

Manufacturing is outsourced to external partners, while the company is headquartered in Amravati and has operational support in Pune. This structure allows a stronger focus on R&D, IP creation, and system design rather than capital-intensive manufacturing infrastructure.

Despite early traction, scaling

remains the company's biggest challenge. Building a stronger ecosystem of OEM collaborators, distribution partners, and strategic investors is critical to transitioning from pilot deployments to large-scale adoption. The startup is also exploring partnerships with semiconductor manufacturers and energy-focused enterprises to strengthen its supply chain and deployment capabilities.

Looking ahead, the startup plans to develop AI- and IoT-enabled energy optimisation systems aimed at predictive power management and carbon credit integration. The company is actively seeking channel partners, semiconductor vendors, and strategic investors to accelerate expansion across industrial, commercial, and consumer markets.

"Innovation is challenging at every stage, from zero to one and then from one to ten," the co-founder notes. "Our goal is to turn energy waste into a measurable resource and demonstrate that India can lead in building scalable deep-tech solutions for sustainability."

With dual patented technology, early commercial deployments, and a roadmap towards intelligent energy systems, Inno 91 Green Technologies is positioning itself as an emerging player in India's deep-tech sustainability landscape, targeting one of the most persistent inefficiencies in modern electricity consumption.

—Saba Aafreen

3 DURABLE, LIGHTWEIGHT, CORROSION-RESISTANT POLES FROM CNC TECHNICS

Helipole is a fibre-reinforced composite pole designed for applications such as street lighting, telecom towers, smart poles, and electrical distribution.

Helipole, a brand developed by CNC Technics Pvt Ltd, was launched in 2018 by Vijay Krishna Bandaru along with

his sons Rahul and Rohit. The company found that traditional steel-and-concrete poles were prone to failure, particularly in

coastal and high-wind regions. Drawing on their expertise in composite materials, CNC Technics (incorporated in 1986) introduced